

Source Weyl energy, refuted as a decay

Exact Dirichlet representation of the centered mass difference, the CD-transfer norm, and why E^{bulk} does not tend to zero

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August 29, 2026

Abstract

After the high-moment kill failed (r398), the reviewer route to $R_k^\dagger \succ \frac{1}{2}I$ on the selected sequence runs through a source-pure Weyl energy of the centered mass difference $d = \mu - \nu - d^{\text{ref}}$. This note records the exact Fourier–Dirichlet representation of $\widehat{d}_k(r)$ (tent interpolant, even-circulant extension, Fejér Laplacian, arch/Gamma/pole subtracted as d^{ref}), derives the transfer weights $\omega_{k,r}$ from the r389 Chebyshev–Christoffel–Darboux identity, splits the three source-defined edge modes, and attacks the bulk sum by Montgomery–Vaughan mean value and Gallagher’s lemma on the lag lattice. What is proved: the representation, the Laplacian identity, $C_0 = C_1 = 0$ identically after Fejér, and that the CD band $m < N_w - 3$ already excludes the cap. What is refuted: $E_k^{\text{bulk}} \leq \varepsilon_k \rightarrow 0$ in that norm. On core-42 the energy grows as $N_w^{0.54}$ and sits at quadratic-mean (mean $C_m^2/\text{qm} \in [0.43, 0.85]$); on the selected sequence $a_k = 2^k$ it grows through $k = 9$. The unconditional large-sieve majorant is 10^3 – 10^6 too large. A bound that would force decay below quadratic-mean is RH-near, hence circular as a strategy for $R^\dagger \succ \frac{1}{2}I$, and is in any case contradicted by the census. No RH claim.

Status. Lemmas 1–4 are proved (finite identities). Lemma 3 names the transfer norm. Proposition 1 is a finite machine census. Proposition 2 is the honesty-gate adjudication. No RH claim. No L^* claim. No $R^\dagger$ claim.

1 The object

On a window of log-cutoff α and lag-mesh Δ , write M for the number of lags, $L = 2M - 2$ for the circulant length, $\theta_j = 2\pi j/L$. The prime lag sequence is the tent interpolant of the von Mangoldt comb [1, 2]

$$c_i^P = - \sum_{n \leq e^{2\alpha}} \frac{\Lambda(n)}{\sqrt{n}} t_\Delta^{\text{circ}}(i\Delta - \log n), \quad (1)$$

t_Δ the unit tent of width Δ , even-extended across $i \mapsto L-i$, plus the reflected tent for $\log n < \Delta$. The arch lags c^A are the explicit incomplete-Gamma / digamma kernel of [1] (Euler, $\log \pi$, Bose factor $e^{-w/2}/(1 - e^{-2w})$). Spectral densities $d^P = \text{FFT}(c_{\text{ext}}^P)$ and $d^A = \text{FFT}(c_{\text{ext}}^A)$ are real and even. Folding $j \leftrightarrow L-j$ onto $x = \cos \theta_j$ with the Chebyshev- U factor $(2/L)(1 - \cos \theta)$ produces the signed union $\mu - \nu$.

Definition 1 (centered mass). *On the folded cosine grid $j = 1, \dots, L/2$,*

$$d_{k,j} = \mu_{k,j} - \nu_{k,j} - d_{k,j}^{\text{ref}} = \frac{2}{L}(1 - \cos \theta_j) d_j^P,$$

where $d_j^{\text{ref}} = (2/L)(1 - \cos \theta_j) d_j^A$ is the explicit Bernstein–Szegő / arch background of [4] (Fejér on the full cosine grid is discrete Chebyshev- U). The last node $j = L/2$ carries the fold-half.

The Fourier coefficients on the unfolded circle are $\widehat{d}_k(r) = \sum_{j=0}^{L-1} d_{k,j}^{\text{unf}} e^{-2\pi i r j/L}$, and the folded cosine transform is $C_m = \sum_{j=1}^{L/2} d_{k,j} \cos(m\theta_j)$. Machine identity: $C_m = \frac{1}{2} \text{Re } \widehat{d}_k(m)$.

2 Exact representation

Lemma 1 (tent interpolant). $d_{k,j} = (2/L)(1 - \cos \theta_j) d_j^P$ identically (folded maxdev 0 on FRAME-A $w = 9$). For every frequency bin j ,

$$d_j^P = - \sum_{n \leq e^{2\alpha}} \frac{\Lambda(n)}{\sqrt{n}} K_{k,j}(\log n),$$

where $K_{k,j}(u)$ is the discrete Fourier transform at bin j of the even-circulant tent centered at u . Writing $t_{k,j} = 2\pi j/(L\Delta)$ and $f_n = \{\log n/\Delta\}$, $K_{k,j}(\log n) = n^{-it_{k,j}} \sigma_k(f_n, j)$ with interpolant symbol $|\sigma_k| \leq 1$ (two-tap plus reflection: a contractive linear operator). The representation is the Dirichlet polynomial $S_k(t) = \sum_{n \leq e^{2\alpha}} \Lambda(n) n^{-1/2+it}$ twisted by σ_k , plus the explicit arch/Gamma/pole terms that have been subtracted as d^{ref} . Checked at machine precision on $w = 9$ (maxdev $2.5 \cdot 10^{-13}$), $kz = 18$ ($2.0 \cdot 10^{-13}$) and $kz = 52$ ($3.9 \cdot 10^{-12}$).

The tent width equals the mesh, so $K_{k,j}$ has no extra sinc-decay in j : high-frequency decay, if any, must come from cancellation in $S_k(t_{k,j})$ itself. Folding is the even extension, also linear and contractive.

Lemma 2 (Fejér Laplacian). Let a be the even-circulant extension of c^P and $\Delta a = 2a - \text{roll}(a, 1) - \text{roll}(a, -1)$. On the unfolded grid,

$$\text{IFFT}(\text{Fej} \odot d^P) = \frac{\Delta a}{L}, \quad \text{Fej}_j = \frac{2}{L}(1 - \cos \theta_j).$$

The stencil is $h_0 = 2/L$, $h_{\pm 1} = -1/L$, mass zero. Exact over $\mathbb{Q}$ on a 6-cycle; FRAME-A residual $8.7 \cdot 10^{-19}$.

Thus $\widehat{d}_k(m)$ is (up to the conventional $\frac{1}{2}$) the m -th Fourier coefficient of the discrete Laplacian of the tent-sampled von Mangoldt comb: a Dirichlet polynomial in the lags.

3 The transfer norm

Lemma 3 (CD weights). The r389 Chebyshev- T Christoffel–Darboux energy of a signed atomic mass v on the cosine grid is

$$\Phi_k = \frac{1}{\pi} C_0^2 + \frac{2}{\pi} \sum_{m=1}^{k-1} C_m^2, \quad C_m = \sum_j v_j \cos(m\theta_j).$$

Taking $v = d$ and $k = N_w - 3$ (the depth of R_{N-3} and of the Uvarov kernel that feeds $\Delta^2 \log \tau$ [5, 6]) gives the transfer weights $\omega_0 = 1/\pi$, $\omega_m = 2/\pi$ for $1 \leq m < N_w - 3$, and $\omega_m = 0$ outside the CD band. This is the unique bandlimited CD bilinear; it is not a fitted kernel.

The r392 Uvarov identity writes $\gamma_n(\mu)$ as a τ -ratio against the Bernstein–Szegő chain, with $\tau_n = \det(I - K_n[\Xi]W)$ and K_n the CD kernel of the background. The Hilbert–Schmidt energy of that kernel on the deleted nodes is exactly Φ at depth n . Assist and $d\Delta$ are Uvarov readouts of the same Gram. On core-42, $\text{corr}(E/N_w, \text{assist}) = 0.18$: the named norm does *not* dominate Assist as a closing majorant (the N -confound $\text{corr}(E, \text{assist}) = -0.78$ is the opposite slope). The 5.4 bridge is a named energy, not a proved inequality.

4 Edge split

Lemma 4 (three source modes). *The source-defined edge is $\{r = 0\}$ (mass), $\{r = 1, L - 1\}$ (fold fundamental), and $\{r = L/2\}$ (cap / Nyquist). After Definition 1, $C_0 = C_1 = 0$ at machine zero (Fejér vanishes at $\theta = 0$ and the even extension kills the first cosine of a mean-zero density). The cap $m = L/2$ lies outside the CD band $m < N_w - 3$ ($N_w = L/4 + O(1)$). Consequently $E_k^{\text{bulk}} = E_k$ identically: the three modes are not majorized, they are absent from the transfer norm.*

This is the reviewer instruction (low frequencies into the edge model, not a triangle bound) executed source-purely: no eigenvalue peeking, no margin.

5 Census: the energy does not fall

Write $E_k = \Phi_{N_w-3}(d)$ for the CD energy of centered d , and $\text{qm} = \frac{1}{2} \sum_j d_j^2$.

Proposition 1 (census). *On the sealed constructors:*

1. FRAME-A $w = 9$: $E = 1.80686$, $\text{mean } C_m^2/\text{qm} = 0.619$, $C_{\max} = 0.583$ at $m = 129$.
2. $kz = 18$: $E = 1.903$; $kz = 52$: $E = 5.698$.
3. Core-42: $E \in [1.178, 5.983]$, $d \log E / d \log N_w = +0.542$ (grows), E/N_w slope -0.458 , $\text{mean } C_m^2/\text{qm} \in [0.427, 0.848]$.
4. Selected sequence $a_k = 2^k$ of [7] ($m_k = k \cdot 2^{\lfloor \sqrt{k} \rfloor} - 1$, $n \leq a_k^2$, table cap $k \leq 9$): $E^{\text{bulk}} = 0.033, 0.191, 0.384, 0.402, 0.508, 0.536, 0.869$ for $k = 3, \dots, 9$. Not monotone to 0. ($k = 2$ is all-edge.)

The bulk sits at quadratic-mean, the same Parseval class as the r389 ν -angles. It does not tend to 0 in the CD-transfer norm.

6 Bound inventory and honesty gate

The frequencies $t_{k,r} = 2\pi r/(L\Delta)$ are an arithmetic progression of spacing $\delta = 2\pi/(L\Delta)$: a large-sieve configuration, not a single- t problem. Prime powers $n = p^\ell$, $\ell \geq 2$, contribute a lag- L^2 ratio 0.23 on FRAME-A; the series $\sum_{\ell \geq 2} p^{-\ell/2} \log p$ converges absolutely (elementary).

block	source	class	role
tent / fold / Fejér Laplacian	Lemmas 1,2	unconditional, finite	exact $\hat{d}$
arch / Gamma / pole as d^{ref}	v563 $A(s)$	unconditional, explicit	centering
CD weights ω	r389	unconditional, finite	transfer norm
p^ℓ , $\ell \geq 2$	Euler product	unconditional, elementary	absolute tail
Montgomery–Vaughan MVT	$\int_0^T S(t) ^2 dt \ll (T+X) \sum a_n ^2$	unconditional (1973)	mean square
Gallagher on the grid	spacing δ	unconditional	$\sum_r S(t_r) ^2$
zero-free / Vinogradov $\psi - x$	classical ZF	PNT-level	not sub-QM
$ S(\frac{1}{2} + it) \ll t^\varepsilon$	critical line	RH-near	would be needed for $E -$

On FRAME-A the crude majorant $(1 + X/T) (\sum \Lambda(n)^2/n) \cdot k$ exceeds E by $3 \cdot 10^3$; at $N_w = 878$ the ratio is $4 \cdot 10^5$ ($X/T = 53$). The unconditional large-sieve exponent lives at or above $(1 + X/T)$ times Parseval, which grows. The data sit at Parseval. Decay of E^{bulk} would require a sub-quadratic-mean bound on $S_k(t_{k,r})$ along the lag lattice — quality of ζ'/ζ on the critical line, or equivalent — which as an input to $R^\dagger \succ \frac{1}{2}I$ is circular (Weil positivity along the selected windows is the target).

Proposition 2 (honesty gate). *The needed large-sieve exponent is not in the unconditional range. A bound that closed $E^{\text{bulk}} \rightarrow 0$ would be RH-near, hence circular as a route to the dual-resolvent contraction. Independently, the finite census of Proposition 1 refutes the decay in the named norm.*

Vaughan’s identity and a Heath–Brown decomposition of Λ were not needed: already the mean-value theorem at classical quality fails to close, and the census is at the random scale those theorems predict, not below it.

7 Kills

- Proposition 3** (controls). *1. Scramble (seed 1, log-positions uniform in $(0, 2\alpha)$, weights kept): $E^{\text{bulk}} = 3.39 > 1.2$ times MAIN 1.81. The centered difference does separate scramble, unlike the r389 ν -angle quadratic-mean (which scramble held). The cancellation that MAIN spends lives in the μ/lag structure, not in the ν -angles.*
- 2. Two-period on the cosine grid, $S = 21$, $c = 2/3$: $\text{HHI} = 0.629$, comb at $m = S$. The concentrating adversary is a comb, as in r389; $\rho_{\text{AP}} = 1$ remains spectral concentration on a progression.*
- 3. Mutant $\omega \equiv 1$ (drop $2/\pi$ and the CD cutoff): $\sum C_m^2 = 2.85$ against $\Phi = 1.81$ — the weights are a scale, not a decay mechanism. Dropping the already-vanishing modes $m = 0, 1$ does not change Φ : the “energy must not fall without the split” mutant is vacuous at this norm and is recorded as such.*
- 4. Dead χ_3 -15: $E = 1.69$; living χ_3 -9: $E = 1.47$; MAIN: 1.81. Two-sided: both $O(1)$. The test is not a world separator (the same honesty as r398).*

8 What remains

The representation and the transfer norm are available as finite theorems for r400 (bulk one-defect / frame inequality). They do not supply a vanishing bulk energy. The macroscopic 1/2-cluster of R_{N-3} (r398: 36 eigenvalues in $[\frac{1}{2}, \frac{3}{5}]$ on FRAME-A) is consistent with a quadratic-mean d whose CD energy grows: a Hilbert–Schmidt smallness that would collapse the cluster is exactly what $E^{\text{bulk}} \rightarrow 0$ would have been, and is not observed. The edge model must absorb more than three modes, or the frame inequality of 5.4 must treat the 1/2-cluster collectively, without asking the Weyl energy to vanish.

Machine check

Standalone: Fractions stencil and Laplacian, energy-path firewall. Construction: FRAME-A representation, Laplacian, $C_0 = C_1 = 0$, E -pin, two further windows, scramble, two-period, dead and living χ , selected $k = 4 < k = 6$, MVT ratio. Discovery census: `source_weyl_energy_probe.py` (`--smoke 22/22 / full 29/29`), core-42, selected $a_k = 2^k$. Exit line: SOURCE WEYL VERIFIED.

Claim boundary

Finite identities for the Dirichlet interpolant, the Fejér Laplacian, and the CD-transfer energy of a centered mass; a named census that this energy does not tend to zero and sits at quadratic-mean; a named honesty-gate that an analytic closing bound would be circular. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

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